

HW4

```
library(stats)
```

Problem 4

Let $N \sim \text{Pois}(\lambda_1)$ be the number of movies that will be released next year. Suppose that for each movie the number of tickets sold is $\text{Pois}(\lambda_2)$, independently.

Use simulations in R (the statistical programming language) to numerically estimate mean and the variance of the number of movie tickets that will be sold next year assuming that the mean number of movies released each year in the US is 700, and that, on average, 800000 tickets were sold for each movie

```
# estimate the average number of movie tickets sold which we know is:
```

```
tickets_sold = rpois(n = 100000, lambda = 800)
movies_released = rpois(n = 10000, lambda = 700)

mean(tickets_sold*movies_released)
```

```
## [1] 560284.5
```